

Question (4): Let $v, w \in \mathbb{R}^n$ be two non-zero vectors.

We're working with the **identity matrix** I_n and matrices B and C , defined as follows:

- $B = I_n - 2(vv^T)$,
- $C = I_n - 2(ww^T)$, where $\|w\| = 1$ (the length of w is 1).

We need to:

1. Check if B is **orthogonal** or **symmetric**.
 2. Check if C is **orthogonal**.
 3. Show that the rank of $I_n - vv^T$ is less than n when $\|v\| = 1$.
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Key Concepts

Let's break down the terms to make them clear:

1. Identity Matrix (I_n):

- This is a special matrix where all diagonal elements are 1 and everything else is 0. For example:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- When you multiply I_n by any vector, the vector stays the same.

2. Orthogonal Matrix:

- A matrix is **orthogonal** if multiplying it by its transpose gives back the identity matrix:

$$B^T B = I_n.$$

- This means the matrix preserves the length and angles of vectors.

3. Symmetric Matrix:

- A matrix is **symmetric** if flipping it along the diagonal doesn't change it:

$$B^T = B.$$

4. Transpose (vv^T):

- vv^T is called the **outer product**. It's formed when a vector v is multiplied by itself, like this:

$$vv^T = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} v_1^2 & v_1 v_2 \\ v_2 v_1 & v_2^2 \end{bmatrix}.$$

5. Euclidean Norm ($\|v\|$):

- The Euclidean norm is the "length" of a vector:

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

6. Rank of a Matrix:

- The rank tells us how many independent rows or columns a matrix has. For example:
 - I_n has full rank (n) because all rows/columns are independent.
 - vv^T has rank 1 because it is built from a single vector.

(a) Is $B = I_n - 2(vv^T)$ Orthogonal and Symmetric?

Step 1: Check if B is Symmetric

1. Start with the definition of B :

$$B = I_n - 2(vv^T).$$

2. To check symmetry, we compute the transpose B^T :

$$B^T = (I_n - 2vv^T)^T.$$

3. Recall these properties:

- The transpose of I_n is itself ($I_n^T = I_n$).
- The transpose of vv^T is also itself ($(vv^T)^T = vv^T$), because $v^T v$ is symmetric.

4. Substituting these:

$$B^T = I_n - 2vv^T.$$

5. Since $B^T = B$, B is symmetric.

Conclusion: B is symmetric for any v .

Step 2: Check if B is Orthogonal

1. To check orthogonality, we calculate $B^T B$:

$$B^T B = BB = (I_n - 2vv^T)(I_n - 2vv^T).$$

2. Expand this product:

$$BB = I_n - 2vv^T - 2vv^T + 4(vv^T)(vv^T).$$

3. Simplify:

- $vv^T vv^T = v(v^T v)v^T$, where $v^T v = \|v\|^2$ is the square of the length of v .

- Let $\|v\|^2 = c$. Then:

$$BB = I_n - 4vv^T + 4c vv^T.$$

4. For B to be orthogonal, $BB = I_n$. But if $\|v\|^2 \neq 1$, $BB \neq I_n$.

Conclusion: B is not orthogonal for general v , but if $\|v\| = 1$, it can be orthogonal.

(b) Is $C = I_n - 2(ww^T)$ Orthogonal?

Here, $\|w\| = 1$, meaning w is a unit vector.

Step 1: Compute $C^T C$

1. Start with:

$$C = I_n - 2ww^T.$$

2. Compute $C^T C$:

$$C^T C = CC = (I_n - 2ww^T)(I_n - 2ww^T).$$

3. Expand the product:

$$CC = I_n - 2ww^T - 2ww^T + 4(ww^T)(ww^T).$$

4. Simplify:

- Since $\|w\| = 1$, $w^T w = 1$, so:

$$(ww^T)(ww^T) = ww^T.$$

Substituting back:

$$CC = I_n - 4ww^T + 4ww^T = I_n.$$

Conclusion: C is orthogonal when $\|w\| = 1$.

(c) Show the Rank of $I_n - vv^T$ is Less Than n When $\|v\| = 1$

Step 1: Analyze $I_n - vv^T$

1. Start with:

$$I_n - vv^T.$$

2. vv^T is a rank-1 matrix because it's formed from a single vector v . The rank of a matrix tells us how many independent rows or columns it has:
- vv^T has rank 1 (it depends only on v).

Step 2: Subtracting vv^T from I_n

1. The identity matrix I_n has full rank n .
2. Subtracting vv^T removes 1 degree of freedom because vv^T reduces the dimension of the space by 1. This happens because vv^T projects vectors onto v , leaving a subspace of dimension $n - 1$ (all vectors orthogonal to v).

3. Therefore:

$$\text{Rank}(I_n - vv^T) = n - 1.$$

Conclusion: The rank of $I_n - vv^T$ is $n - 1$, which is less than n .

Final Answers

1. (a):

- B is symmetric for any v .
- B is not orthogonal for general v .

2. (b):

- C is orthogonal when $\|w\| = 1$.

3. (c):

- The rank of $I_n - vv^T$ is $n - 1$ when $\|v\| = 1$.